.NET Fundamentals



Soyeb Ghachi

Corporate Trainer | Mentor |SME

Overview



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History of .Net

- Sometime in the July 2000, Microsoft announced a whole new software development framework for Windows called .NET in the Professional Developer Conference (PDC).
- Microsoft started development on the .NET Framework in under the name of Next Generation Windows Services (NGWS). By late 2000 the first beta versions of .NET 1.0 were released.

.Net SDK Releases

.NET SDK's (Software Development Kit)

IDE (Integrated Dev. Environment)	.Net Framework	.Net Core
VS.NET 2002	1.0	N/A
VS.NET 2003	1.1	N/A
VS.NET 2005	2.0	N/A
VS.NET 2008	2.0, 3.0, 3.5	N/A
VS.NET 2010	2.0, 3.0, 3.5, 4.0	N/A
VS.NET 2012,13,14	2.0, 3.0, 3.5, 4.0,4.5	N/A
VS.NET 2015	2.0,3.0.3.5,4.0,4.5,4.6	N/A
VS.NET 2017	2.0,3.0.3.5,4.0,4.5,4.6	.NET Core 1.0,1.1,2.0, 2.2
VS.NET 2019	2.0,3.0.3.5,4.0,4.5,4.6,4.7,4.8	.NET Core 3.0,3.1,3.3
VS.NET 2021	2.0,3.0.3.5,4.0,4.5,4.6,4.7,4.8	.NET 5.0
VS.NET 2022,23	2.0,3.0.3.5,4.0,4.5,4.6,4.7,4.8	.NET 6.0, NET 7.0, .NET 8.0, .NET 9.NET 10

Flavors of .NET

Console Applications:

- These refer to traditional DOS kind of applications like batch scripts.
- It has CUI(Character Based User Interface).

Windows Form Based Applications/UWP (Universal Windows Applications)

- These refer to traditional rich client applications.
- It has GUI (Graphical User Interface)

ASP.NET/ Web applications:

- These include dynamic and data driven browser based applications.

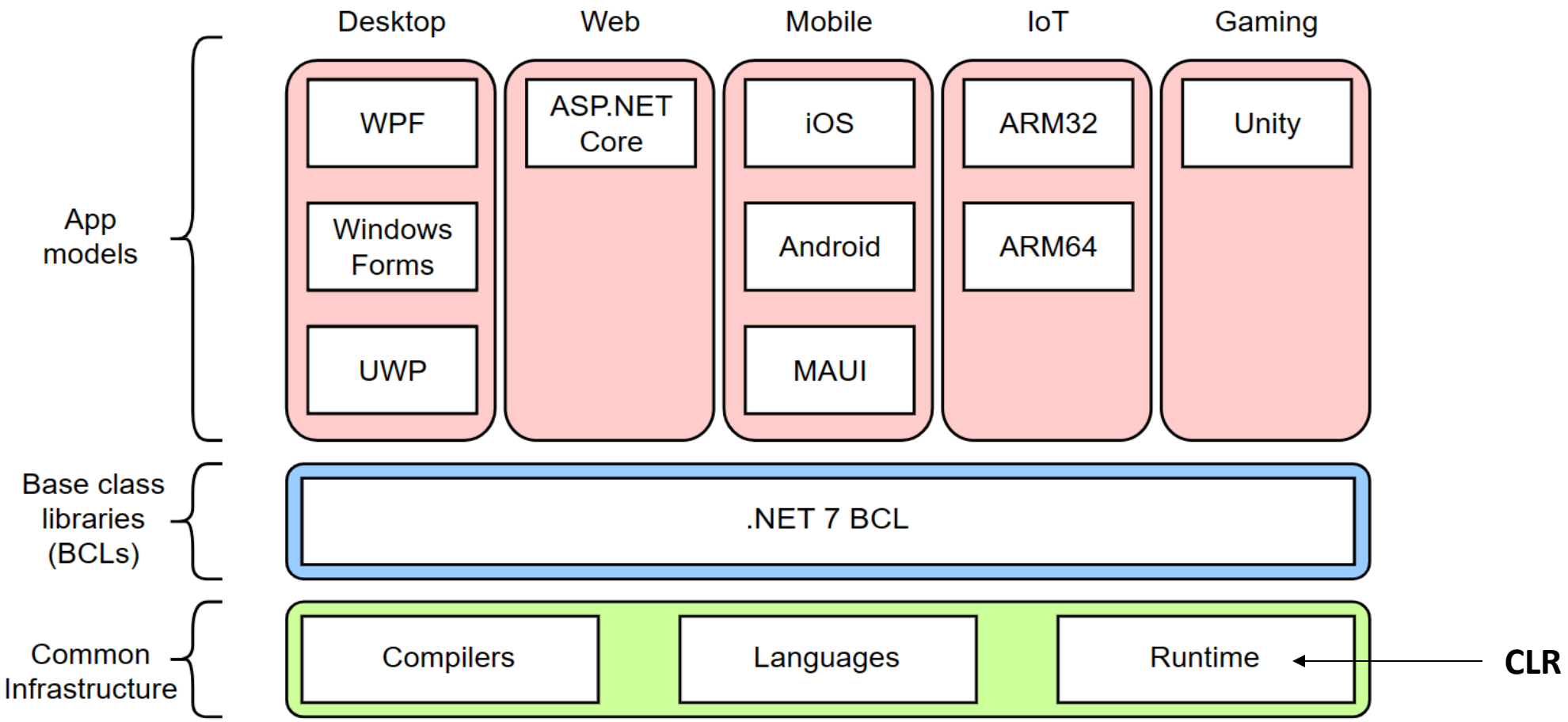
WPF Application (Windows Presentation Foundation):

- WPF is a powerful new infrastructure based on DirectX.
- This means that you can use rich graphical effects without incurring the performance overhead that you'd suffer with Windows Forms.
- In fact, you even get advanced features such as support for video files and 3-D content.

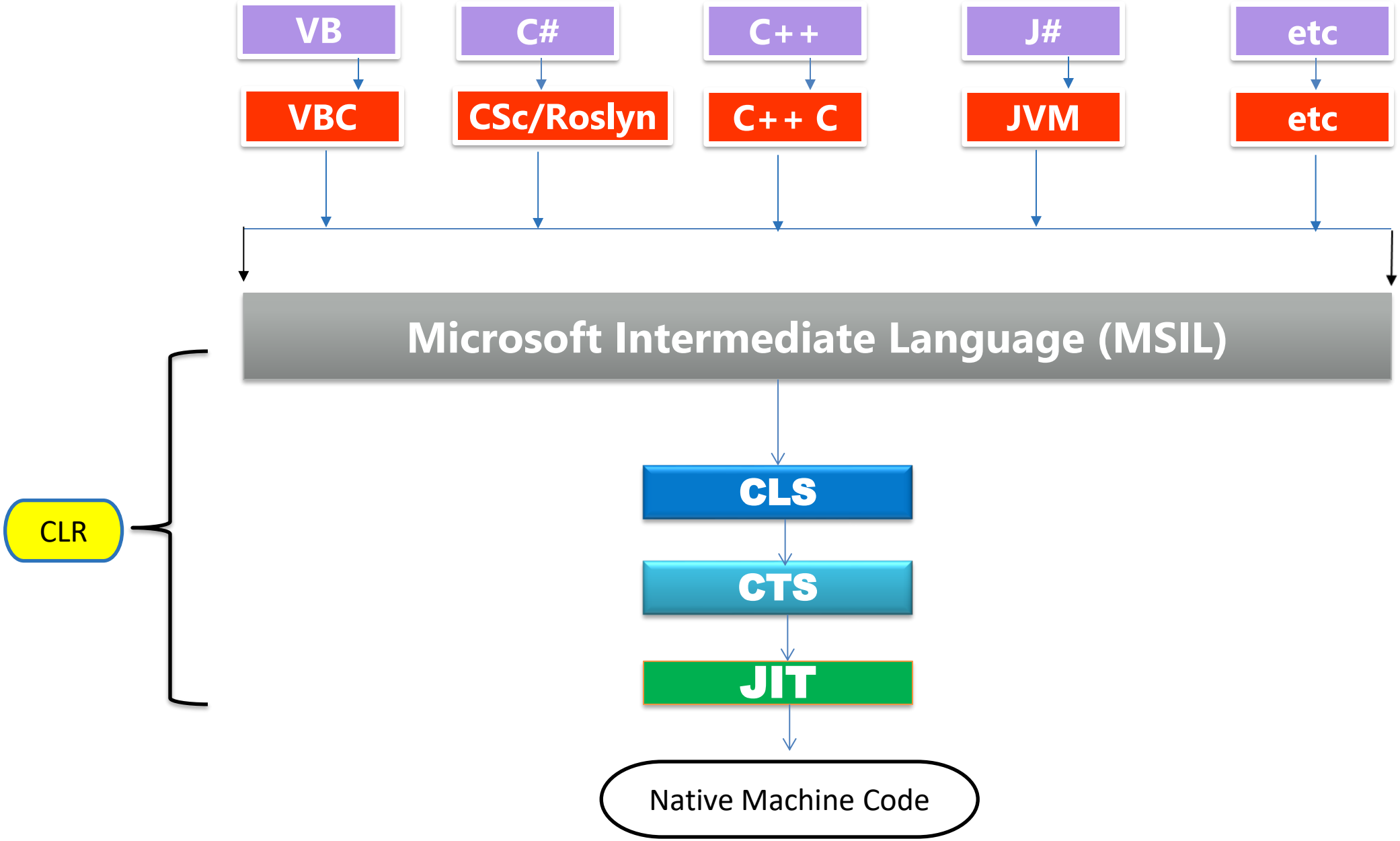
Xamarin/MAUI:

- Its Free, Open Source and Cross platform Mobile App Dev. SDK.
- An app platform for building Android and iOS apps with .NET and C#.

.NET Ecosystem



Virtual Execution System (VES)-CLR


VB
C#
C++
J#
etc
VBC
CSc/Roslyn
C++ C
JVM
etc
Microsoft Intermediate Language (MSIL)
CLR
CLS
CTS
JIT
Native Machine Code

What is the CLR and Role of CLR?

- NET CLR is a run-time environment that manages and executes the code written in any .NET programming language.
- It converts code into native code which further can be executed by the CPU.
- **Base Class Library/Core Library Support**

It is a class library that provides support of classes to the .NET application

- **Type Checker**

It checks types used in the application and verifies that they match to the standards provided by the CLR.

- **Code Manager**

It manages code at execution run-time.

- **Garbage Collector**

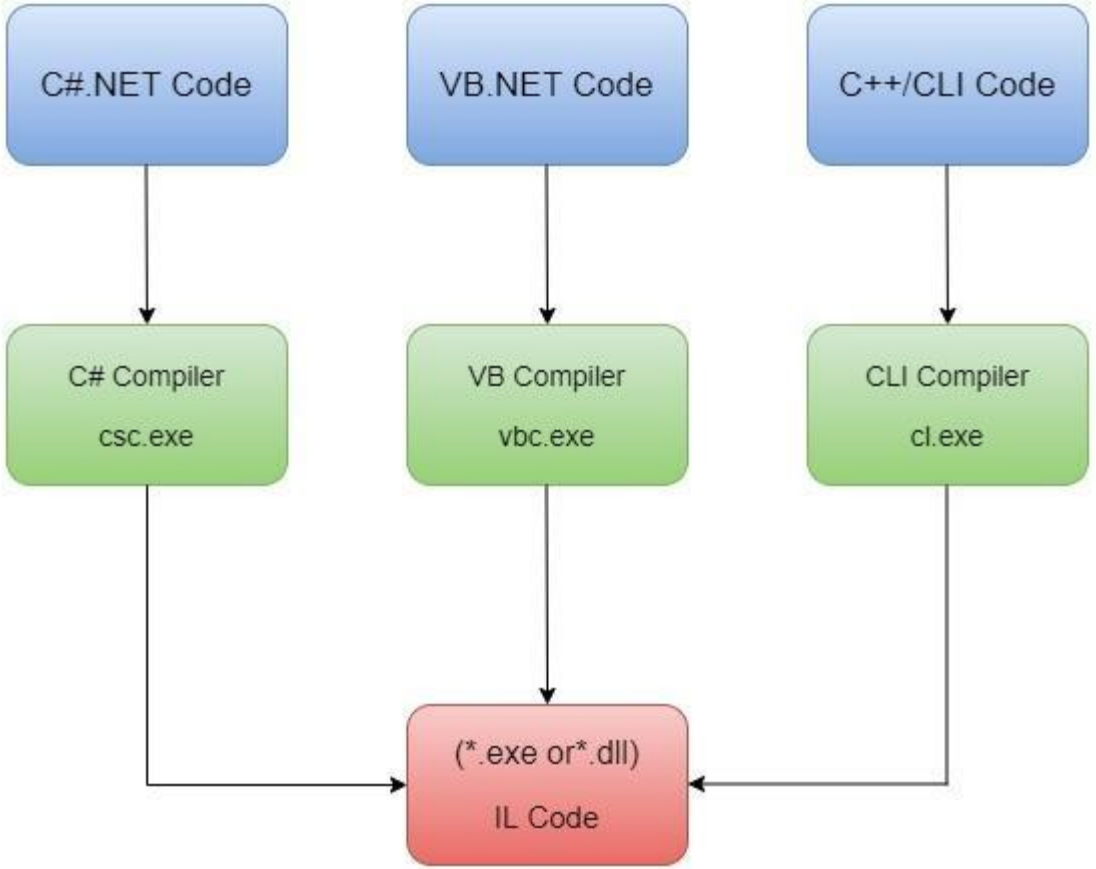
It releases the unused memory and allocates it to a new application.

- **Exception Handler**

It handles the exception at runtime to avoid application failure.

Microsoft Intermediate Language (MSIL)

- is a set of instructions that are platform independent and are generated by the language-specific compiler from the source code.
- The MSIL is platform independent and consequently, it can be executed on any of the Common Language Infrastructure supported environments such as the Windows .NET runtime.



CLS and CTS

CLS (Common Language Specification)

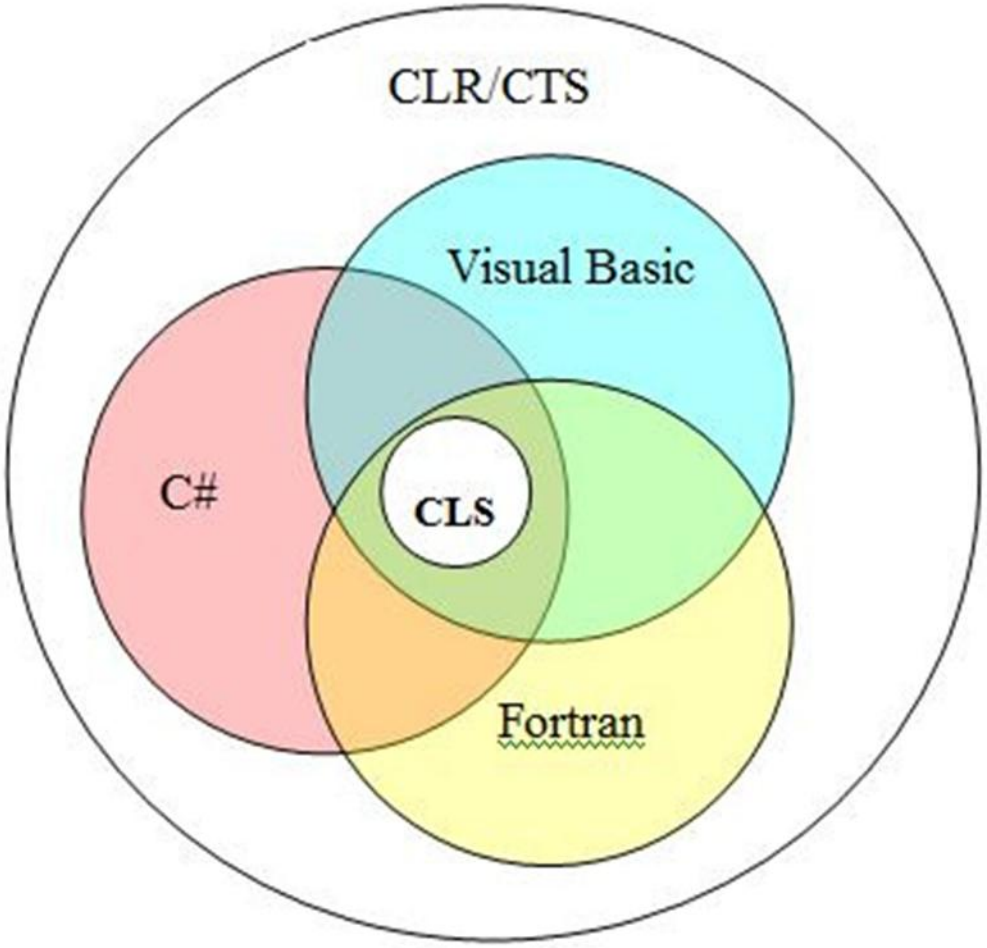
- CLS stands for Common Language Specification and it is a subset of CTS. It defines a set of rules and restrictions that every language must follow which runs under the .NET framework.
- The languages which follow these set of rules are said to be
- CLS Compliant.

For Example

- For each. Net lang we have to use the new keyword to create an instance, index of array of each. Net lang should start with 0.

CTS (Common Type System)

- Common Type System (CTS) describes the datatypes that can be used by managed code. CTS defines how these types are declared, used and managed in the runtime.
- It facilitates cross-language integration, type safety, and high-performance code execution.



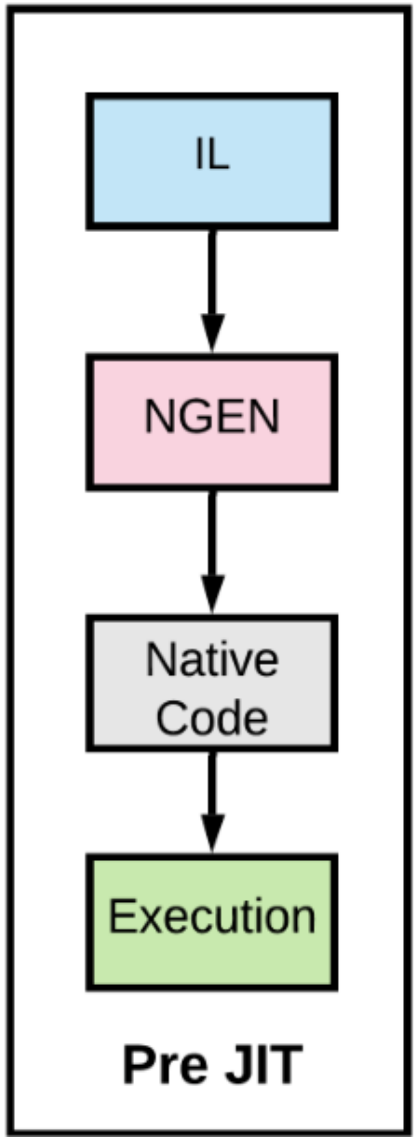
Just-In-Time (JIT)

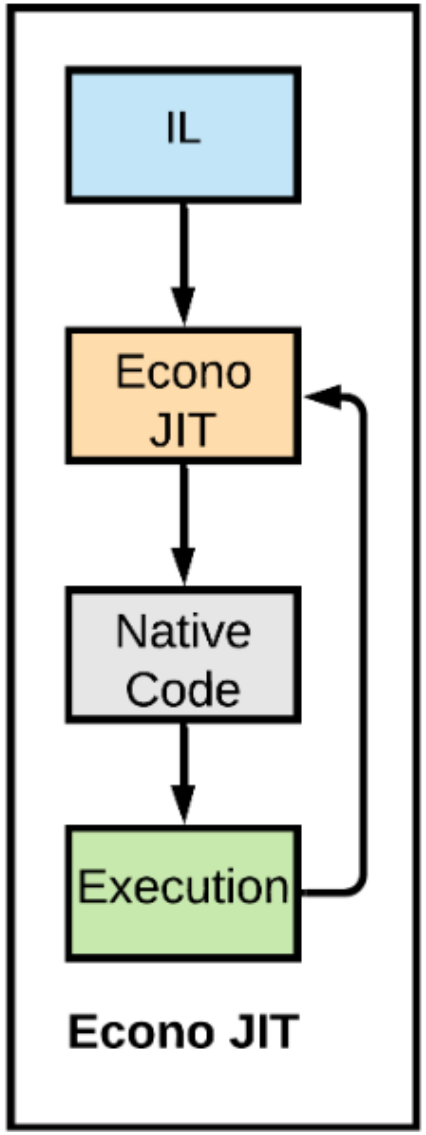
- The JIT compiler translates **the MSIL code of an assembly to native code and uses the CPU architecture of the target machine to execute a . NET application**.

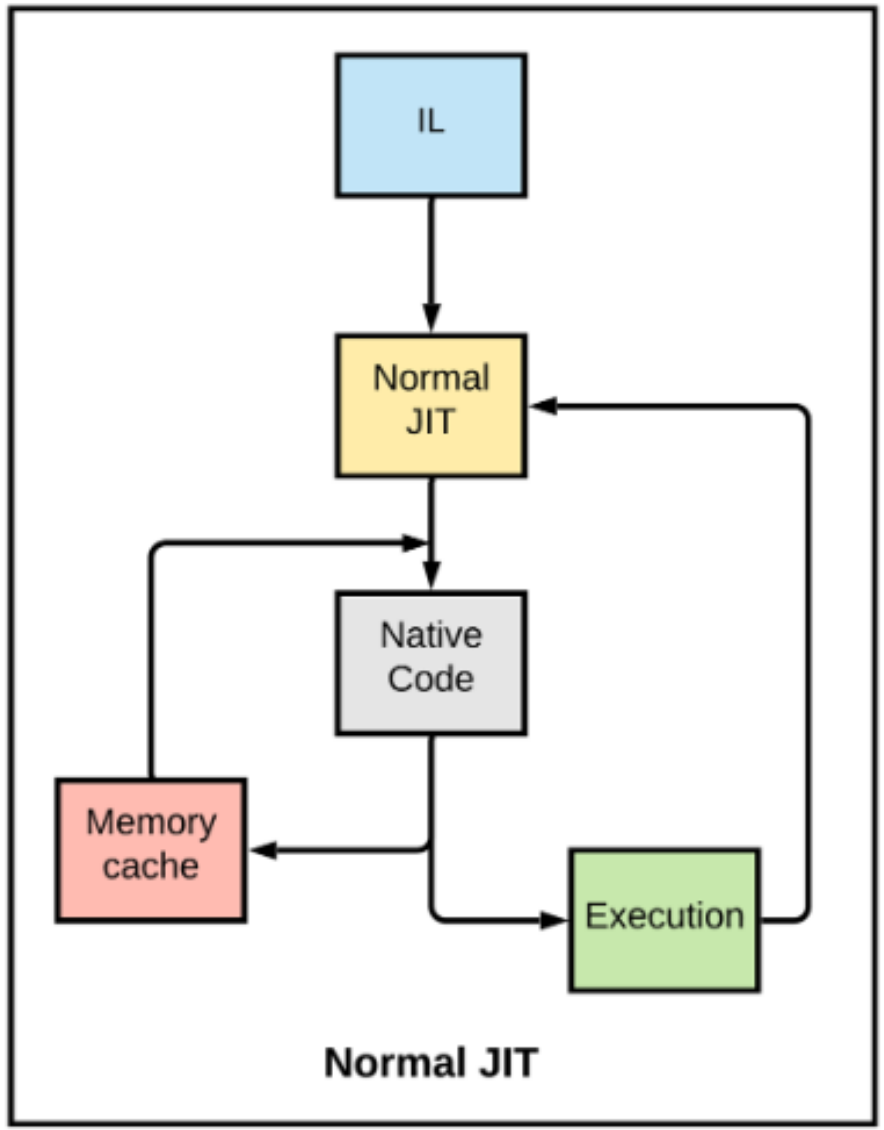
- **Pre-JIT**: Compiles the entire source code during compilation and is used at the time of deployment.

- **Econo-JIT**: Compiles methods that are called during run time.

- **Normal-JIT**: Compiles only the methods called during run time (at the instant of their first call) and stores the compiled code in cache to be used in subsequent calls.


IL
NGEN
Native Code
Execution
Pre JIT


IL
Econo JIT
Native Code
Execution
Econo JIT


IL
Normal JIT
Native Code
Memory cache
Execution
Normal JIT

Major Features of .NET

1. Cross-Platform Development

Modern .NET allows applications to run on multiple operating systems.

Supported platforms

- Windows
- Linux
- macOS
- Android
- IOS

Example

A single ASP.NET Core Web API can run on:

- Windows Server
- Linux Docker Container
- Azure App Service

2. Common Language Runtime (CLR)

CLR is the execution engine of .NET.

3. Language Interoperability

Languages include:

- C#
- VB.NET
- F#
- Python
- Ruby

4. Base Class Library (BCL)

BCL provides a large collection of reusable classes.

Namespaces include:

System

System.IO

System.Collections

System.Data

System.Net

System.Threading

5.Automatic Memory Management (Garbage Collection)

.NET automatically manages memory using **Garbage Collector (GC)**.Developers do **not manually free memory** like C/C++.

.Net Framework and .Net Core

Terms	.Net Fx	.Net Core
Open Source	Proprietary Framework	Open Source Framework
Cross-Platform	.NET Framework enabled developers to build applications for a single platform — Windows.	.NET Core is cross-platform, and supports three distinct operating systems — Windows, OS X, and Linux.
Installation	The .NET Framework needs to be installed as a **single package** and runtime environment for Windows.	.NET Core is cross-platform, and needs to be packaged and installed independent of the underlying operating system.
Modular Collection of Libraries	.NET fx is not cross-platform, and needs to be packaged and installed underlying operating system.	.NET Core is cross-platform, and needs to be packaged and installed independent of the underlying operating system.
Application Models	The application model of .NET Framework includes Windows Forms, ASP.NET, and Windows Presentation Foundation (WPF).	On the other hand, the application model of .NET Core includes ASP.NET Core and Windows Universal Apps.

Deployment Options	.NET Framework, developers have to deploy web applications only on Internet Information Server.	the web applications developed with ASP.NET Core can be hosted in a number of ways.
Cloud Ready Configuration	Required some additional config.	Unlike .NET Framework, .NET Core is designed with features to simplify development and deployment of cloud-based application.
Performance and Scalability	Its comparatively poor than .Net Core App in terms of performance.	NET Core is more effective than .NET Framework to enhance the performance and scalability of applications.
Microservices	Required additional config.	Unlike .NET Framework, .NET Core makes it easier for developers to build microservice oriented systems rapidly.

Features of Visual Studio

Features of Visual Studio-IDE

- Shared IDE
- VS.NET Multi-Targeting Support
- Intellisense
- Strong Debugging Support
- Watch window
- Code Snippet
- Code Test
- Collaboration

Installation Guide: https://learn.microsoft.com/en-us/visualstudio/install/install-visual-studio?view=vs-2022

Introduction to .NET CLI

.NET 8 CLI Commands (Console & ASP.NET Core)

1. Check Installed .NET Version

```
dotnet --version
```

```
dotnet --list-sdks
```

2. Create a Console Application (.NET 8)

```
dotnet new console -n MyConsoleApp --framework net8.0
```

Part	Meaning
dotnet new	Create new project
console	Console template
-n	Project name
net8.0	Target framework

3.Navigate to Project Folder

```
cd MyConsoleApp
```

4.Build Application

```
dotnet build
```

5.Run Application

```
dotnet run
```

Supported Programming Languages in .NET

Visual Basic. Net
C#
APL
4Fortran
Pascal
C++
Haskell
Perl
J#
Python
COBOL
Microsoft JScript

RPG
Component Pascal
Mercury
Scheme
Curriculum
Mondrian
Smalltalk
Eiffel
Oberon
Standard ML
Forth
VC++
F#

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